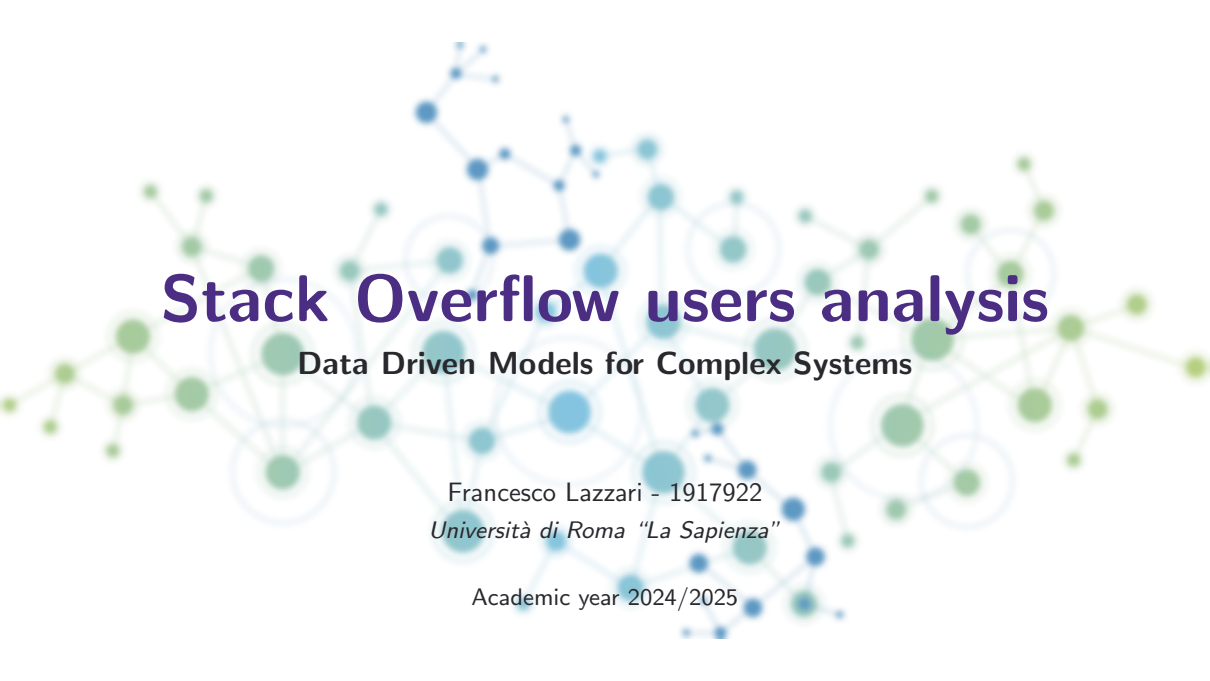




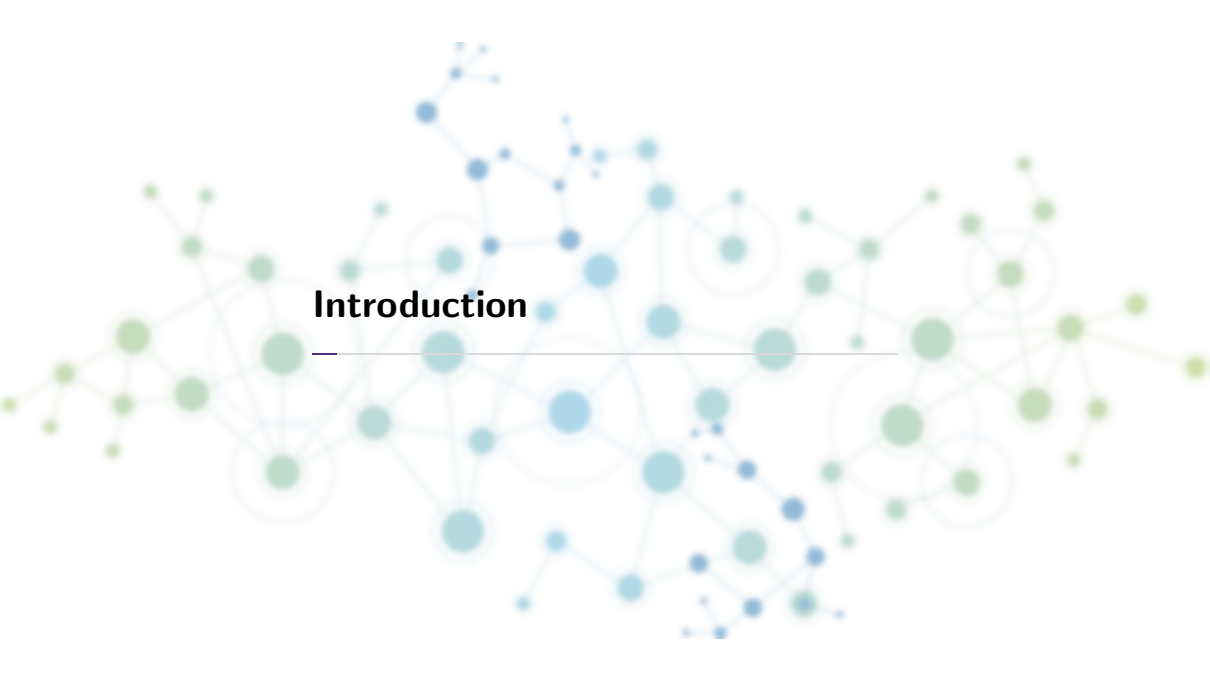
Stack Overflow users analysis

Data Driven Models for Complex Systems

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Introduction

Dataset description

For this project, we aimed to study the interactions among users of the popular platform Stack Overflow and how these interactions have evolved over the years.

To do so, I used the following dataset provided directly by Stack Exchange, freely downloadable at this [link](#), which mainly contains the following data:

- **Post.xml** → a list of all questions and answers posted on the platform for each year (up to 2023), whose topic is summarized by the attached tags.
- **Comments.xml** → a list of all comments posted under each post, specifically identified by a PostId and a UserId.

Given the large size of the data, about 130 GB, it was not possible, due to obvious computational limits, to process them all at once. Therefore, the analyses presented were conducted only on partial data extracted on a yearly basis.



Research Question 1



Do user collaborations form thematically coherent communities?

Users - Questions Bipartite Graph

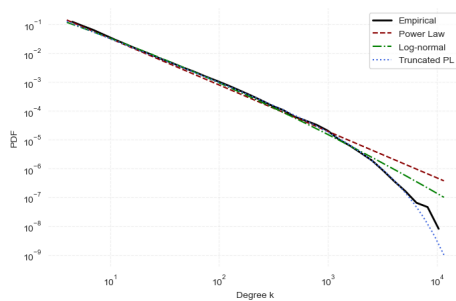
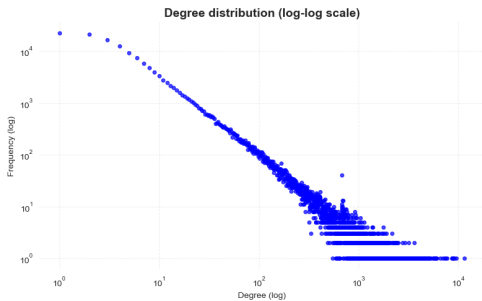
The starting year chosen was 2010, considering the first two years as a time span for the network to grow and gain popularity.

The first step of the analysis was to build a **user-question bipartite graph**, placing an edge whenever a user left a comment or an answer under a question.

The resulting graph was then projected onto the users in order to obtain a **user-collaboration graph**, whose *degree distribution* was first analyzed to gain an initial understanding of the network structure.

Users	Question	Comments
177 387	689 583	2 105 186

User collaboration graph - Degree distribution



α PL	α TPL	λ TPL	$\langle k \rangle$	σ_k	R
1.689	1.492	0.000591	45.8	205.1	-36.7311

From the graphical inspection, we immediately notice that the network in question does not follow a pure power-law trend, as also confirmed by the KS test p-value equal to 0.00.

However, by extending the analysis, we observe that our data follow the trend of a **power law with exponential cutoff**, whose discrete distribution is given by:

$$p_x = x^{-\alpha} e^{-\lambda x}$$

This result is consistent with what is reported in Barabási's book (Chapter 4 – Advanced Topics A–C), where it is explained that networks built from real-world data do not always follow a pure power-law distribution but often exhibit a *crossover distribution*. Moreover, it is noted that the KS test is extremely sensitive, and in the case of very large networks, even small deviations can prevent it from succeeding.

Such a cutoff is also typical of collaboration networks (e.g., actor-collaboration graphs), where structural constraints prevent unbounded growth of node degrees.

Community detection

Given the size of the bipartite graph, we decided to use the **Leiden** algorithm, a modification of the well-known Louvain method, also based on modularity maximization and with a *time complexity* of $O(L)$.

To identify non-overlapping large communities and remove noise, we set a minimum valid community size equal to 1% of the total number of nodes.

Through this filtering process, we identified 7 valid communities, which included 89.2% of the original nodes.

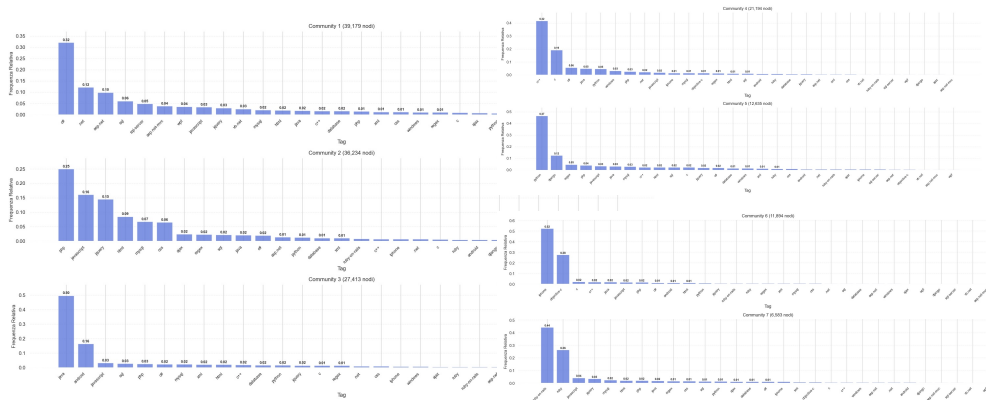
Community tag distribution

From an initial inspection of the tags, we noticed that they included unique values of the order 10^4 . To facilitate the thematic interpretation of the communities, we decided to filter these tags keeping only those that appeared in at least 1% of the questions. Although this filter was relatively mild, it allowed us to reduce the number of tags to 29, highlighting that almost all of them were niche tags.

Finally, to verify the thematic consistency of the 7 large communities, the following steps were carried out:

- extracted the questions commented on or answered by each user;
- counted the frequency of question tags per user;
- grouped the tag frequencies by users according to their community membership, obtaining the tag distribution for each community.

Community tag distribution



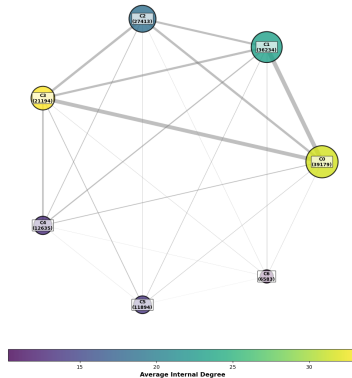
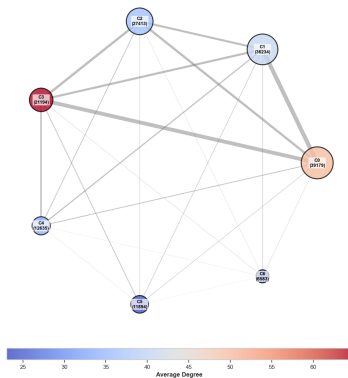
C1	C2	C3	C4	C5	C6	C7
c / SQL	Front-end	java Android app	c++ / c	Python / Django	iOS / obj-c	Ruby

From the inspection of the various tag distributions, we can clearly see that each community groups together users who have collaborated within thematically coherent clusters, thus confirming our initial hypothesis.

To better visualize their interactions, we decided to group the nodes of the user-collaboration graph, where:

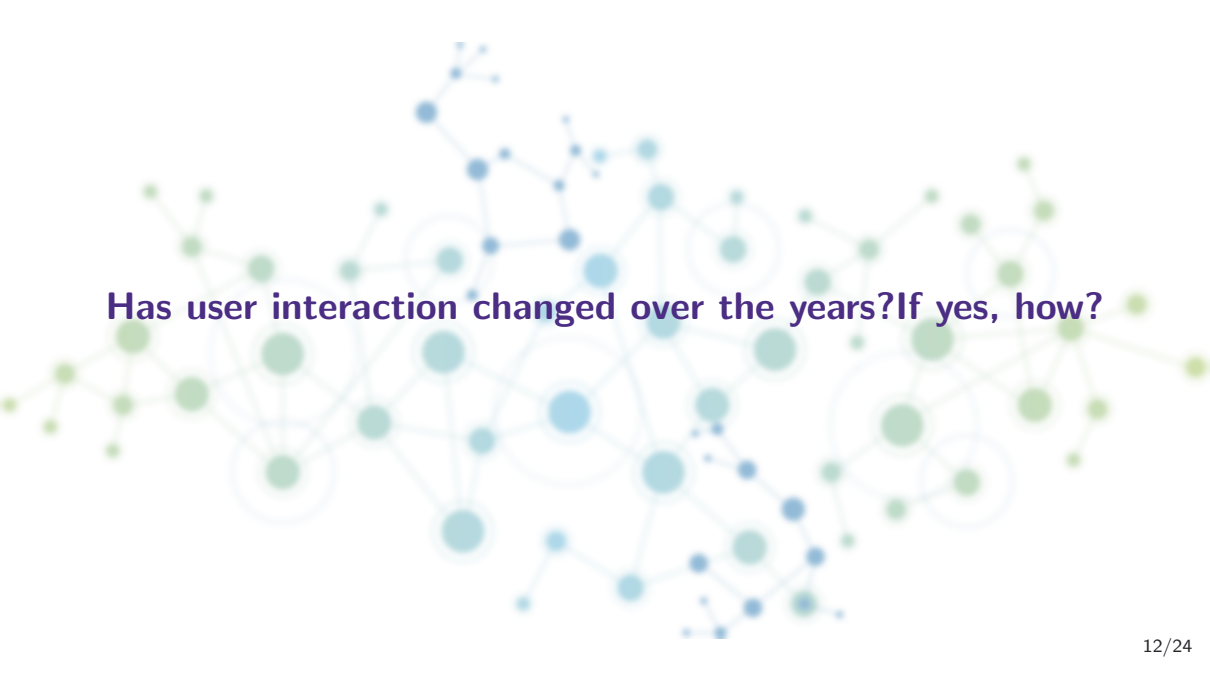
- the size of the nodes indicates the cardinality of the community;
- the thickness of the edges represents the intensity of cross-community collaboration;
- the color indicates the average degree (internal average degree), which reflects the average participation of the users in a community across Stack Overflow (average collaboration within their own community).

User collaboration graph by community





Research Question 2



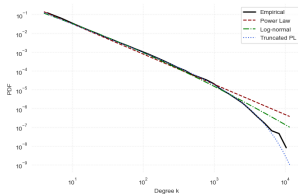
Has user interaction changed over the years? If yes, how?

Network growth

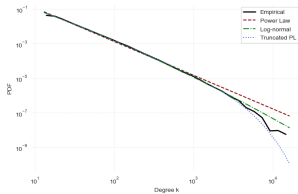
year	Users	Question	Comments
2010	177 387	689 583	2 105 186
2012	464 694	1 619 574	5 668 183
2014	745 324	2 128 799	7 930 476
2016	876 923	2 197 115	8 283 968
2018	874 778	1 885 077	7 013 225
2020	906 657	1 864 330	6 104 948
2022	757 770	1 351 183	4 157 079

From this table, we can already distinguish two phases of the network: an initial growth phase (2010–2016) and a consolidation phase (2018–2022).

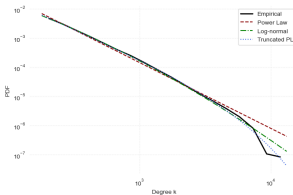
Users collaboration graphs - Degree distribution



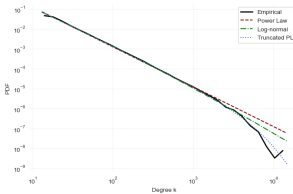
2010



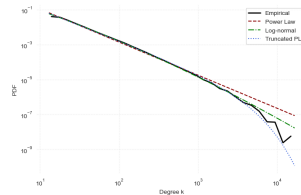
2016



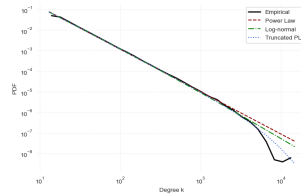
2012



2018

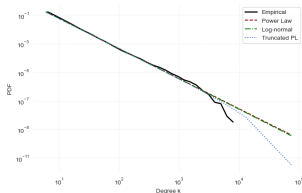


2014



2020

Users collaboration graphs - Degree distribution



year	α PL	α TPL	λ TPL	$\langle k \rangle$	σ_k	R
2010	1.689	1.492	0.000591	45.8	205.1	-36.7311
2012	1.804	1.652	0.000608	30.3	155.9	-36.4717
2014	1.897	1.798	0.000435	22.5	131.8	-29.4990
2016	1.947	1.861	0.000423	18.2	111.3	-25.2534
2018	2.001	1.947	0.000267	14.9	96.4	-19.0214
2020	2.062	2.026	0.000187	11.7	80.9	-13.3230
2022	2.108	2.094	0.000072	9.6	114.2	-4.2364

Note:

To enable a proper comparison of the distribution parameters over the years, all fits were performed using the same value of $x\text{-min} = 13$.

Users collaboration graphs - Degree distribution

Observing the trend of the degree distribution, we can draw the following conclusions:

- Over time, the network stabilizes towards a distribution increasingly close to a canonical power law while still maintaining the presence of an exponential cutoff.
 - Confirmed by the decreasing trend of the Likelihood ratio and the parameter λ .
 - The α parameters increase, reaching a range of values similar to those of other typical collaboration networks with a scale-free distribution.
- The collaboration network becomes progressively sparser, with a reduction in the average number of connections per user. This is an effect of the growth in the number of users not accompanied by a corresponding increase in interaction possibilities (questions), suggesting that collaborations are increasingly concentrated in more specific thematic groups.
- Significant reduction in variance, indicating a distribution less dominated by hubs.

Community description

year	C1	C2	C3	C4	C5
2010	c / SQL	Front-end	java / Android app	c++ / c	Python / Django
2012	Front-end	c / SQL	java	c++ / c	Python / Django
2014	Front-end	c / SQL	c++ / c	java	Android app
2016	Front-end	c / SQL	java	c++ / c	Python / Django
2018	Front-end	c / SQL	Python / Django	java	c++ / c
2020	Front-end	Python	c#	java	c++ / c
2022	Front-end	Python	c#	c++ / c	java

Note:

- From 2012 the community *java / Android app* is split into two dedicated communities with more specific sub-tags.

Community description

year	C6	C7	C8	C9	C10
2010	iOS / obj-c	Ruby			
2012	Android app	iOS / obj-c	Ruby		
2014	Python	iOS / obj-c	Ruby	R	
2016	Android app	iOS / Swift	Ruby	R	
2018	Android app	Backend	iOS / Swift	R	Excel
2020	Android app	AWS / Docker	SQL	iOS / Swift	R
2022	Android app	Backend	SQL	php / laravel	AWS

Note:

- From 2016 the iOS community introduced the *Swift* tag.
- From 2020 the community *c / SQL* split into two dedicated communities.

Community description

year	C11	C12	C13	C14	C15
2020	Excel	php / laravel	react / reactjs javascript		
2022	Excel	R	iOS / Swift	typescript angular	Django Python

Note:

- In 2022 the community *Python / Django* splitted into a pure *Python 3* community and a more dedicated *Django* community with a high presence of Python sub-tags.

Community description

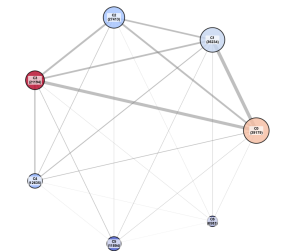
year	number of communities	% nodes in large communities	Valid Tags
2010	7	89.2	29
2012	8	87.3	33
2014	9	87.4	31
2016	9	85.9	34
2018	10	84.9	33
2020	13	81.0	38
2022	15	77.8	40

As hypothesized, over the years, user collaboration has become increasingly thematically specialized, with a progressive increase in communities focused on narrower domains.

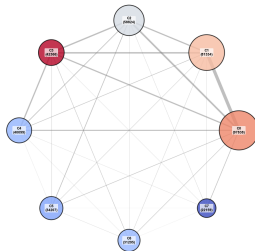
Naturally, the number of communities and the number of valid tags after filtering were found to be strongly correlated.

However, it should be noted that the filter applied to the tags is based on the number of questions (1%), and consequently, the increase in tags is due to the growing popularity of certain topics that were once considered niche

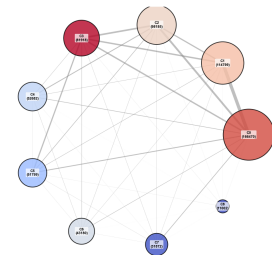
Users collaboration graph by community



2010

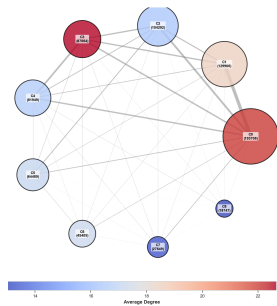


2012

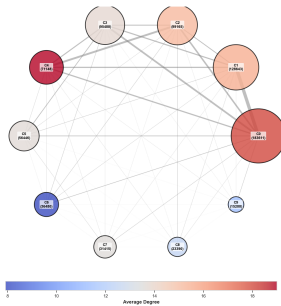


2014

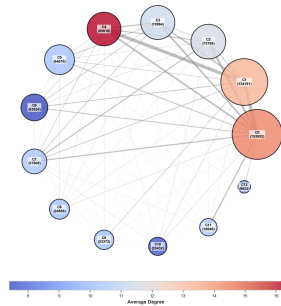
Users collaboration graph by community



2016

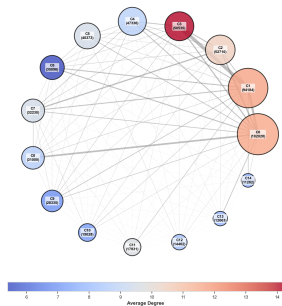


2018



2020

Users collaboration graph by community



2022

year	Global Modularity
2010	0.4117
2012	0.5133
2014	0.5588
2016	0.5912
2018	0.6126
2020	0.6348
2022	0.6441

The progressive structuring of the network is confirmed by the increase in modularity which shows how, over time, the network has divided into increasingly distinct sub-groups. This trend is consistent with the decrease in average degree and the growth in the number of users, which have promoted greater thematic separation between groups.

The decreasing trend in the percentage of nodes within large communities reflects the platform's increasing thematic specialization: a growing number of users belong to smaller but more coherent communities that do not exceed the threshold set by the filter.

Finally, the growth in the number of users was not accompanied by a proportional increase in questions, thereby reducing opportunities for direct interaction between users. Over time, this has led to a decrease in the network's average connectivity and thus in the relative participation (both global and internal) of users, rather than an actual decline in overall activity.



Thanks for your attention
